Least Square Regression

The Daily Grind (Literally)

It is believed that drinking coffee affects productivity and boosts coding skill. To put this theory to the test, a software team tracked the number of coffee cups consumed per day and the number of lines of code written (in thousands) by a programmer over five days. The data collected is as follows:

Day	Coffee Cups	Lines of Code ($\times 1000$)
1	3	12
2	2	10
3	4	14
4	5	4
5	3	11

Table 1: Coffee consumption and coding output over five days.

Answer the following questions based on the given scenario-

i) **Roughly sketch the given data.**

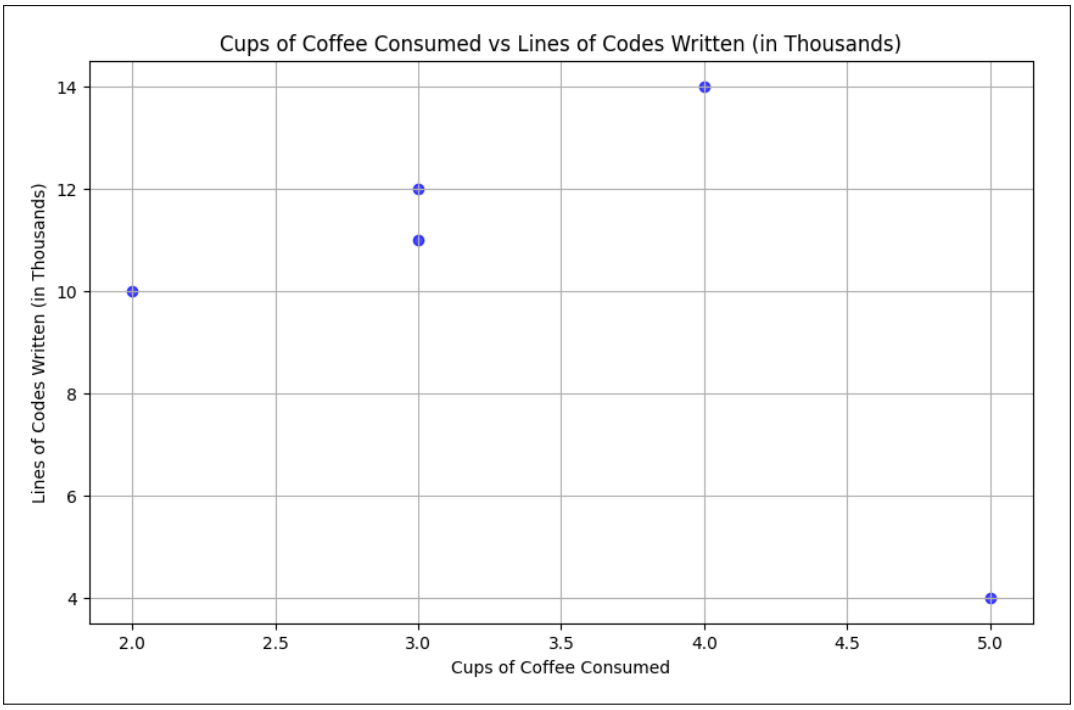

Figure 1: Cups of Coffee Consumed vs Lines of Code Written (in Thousands)

ii) **Identify the outlier from the given points and exclude it from future calculations.**

The point (5, 4) appears to be an outlier. For the rest of the data, productivity generally increases with coffee consumption; however, at this point there is a significant drop in productivity, which does not align with the overall trend. Therefore, this point is more likely an **outlier** or an unwanted noise.

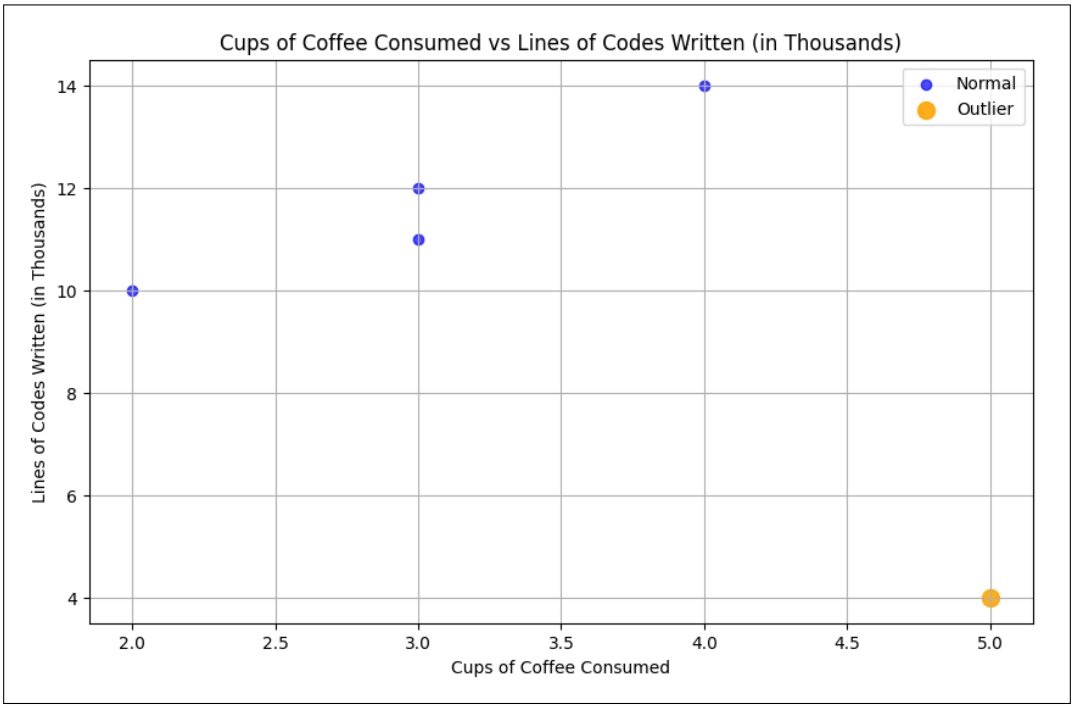

Figure 2: Cups of Coffee Consumed vs Lines of Code Written (in Thousands [with outlier])

So, we will be excluding this point from future calculations.

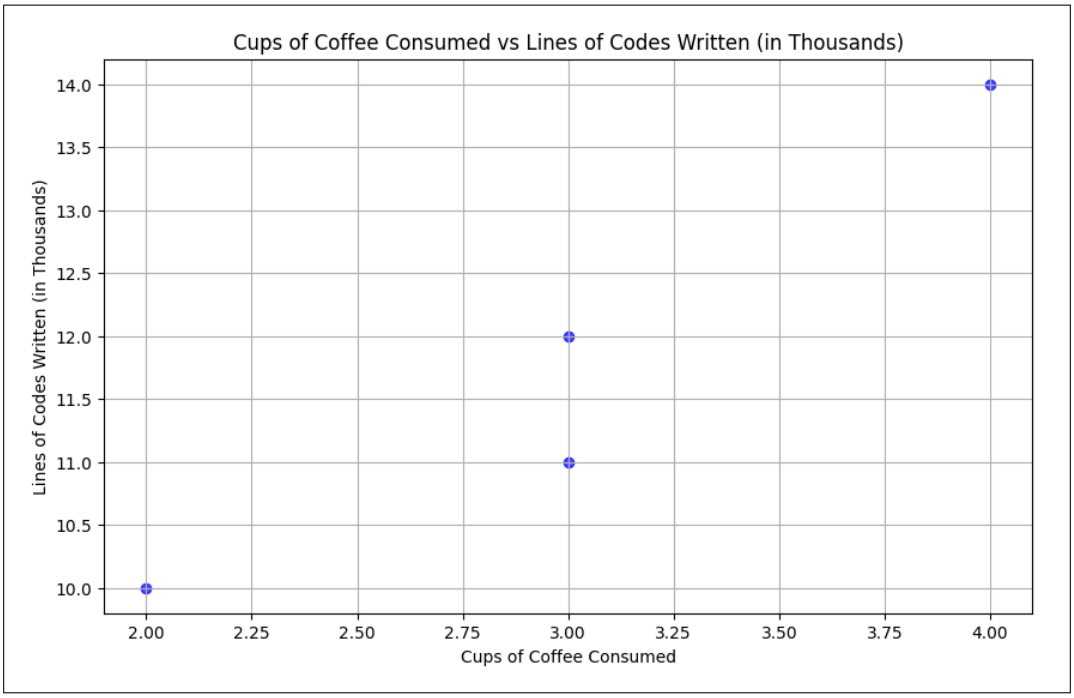

Figure 3: Cups of Coffee Consumed vs Lines of Code Written (in Thousands [without outlier])

iii) **Represent the given data in matrix form** ($Ax = b$)**.** A linear equation is represented as $y = mx + c$ where m is the slope of the equation and c is the y intercept of the line. So, if the given point passes through that line, then

$$3m + c = 12 \tag{1}$$
$$2m + c = 10 \tag{2}$$
$$4m + c = 10 \tag{3}$$
$$3m + c = 11 \tag{4}$$

So,

$$Ax = b$$
$$\implies \begin{bmatrix} 3 & 1 \\ 2 & 1 \\ 4 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 12 \\ 10 \\ 14 \\ 11 \end{bmatrix}$$

But from Figure 3 we see that b does not fall on A so, we can't solve Ax for b, rather we can solve $A\hat{x}$ for p where p is the projection of b on A, in other words the closed thing to b on A. So,

$$A\hat{x} = p$$
$$\implies \begin{bmatrix} 3 & 1 \\ 2 & 1 \\ 4 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = p$$

iv) **It is quite evident that $b \notin C(A)$. So, find the particular solution ($\hat{x}$), projection vector (p), and projection matrix (P) such that**

$$A\hat{x} = p \quad \textbf{and} \quad Pb = p.$$

$$\hat{x} = (A^T A)^{-1} A^T b = \begin{bmatrix} 2 \\ \frac{23}{4} \end{bmatrix}$$

$$p = A\hat{x} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \\ 4 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ \frac{23}{4} \end{bmatrix} = \begin{bmatrix} 47/4 \\ 39/4 \\ 55/4 \\ 47/4 \end{bmatrix}$$

$$P = A(A^T A)^{-1} A^T = \begin{bmatrix} 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 3/4 & -1/4 & 1/4 \\ 1/4 & -1/4 & 3/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \end{bmatrix}$$

v) **Determine the error vector e and verify its correctness.**

$$\text{Error, e} = b - p = \begin{bmatrix} 12 \\ 10 \\ 14 \\ 11 \end{bmatrix} - \begin{bmatrix} 47/4 \\ 39/4 \\ 55/4 \\ 47/4 \end{bmatrix} = \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ -3/4 \end{bmatrix}$$

We know, $p \perp e$, so, $p^T e = \begin{bmatrix} 47/4 \\ 39/4 \\ 55/4 \\ 47/4 \end{bmatrix}^T \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ -3/4 \end{bmatrix} = [0]$

We also know, $A \perp e$, so, $A^T e = \begin{bmatrix} 3 & 1 \\ 2 & 1 \\ 4 & 1 \\ 3 & 1 \end{bmatrix}^T \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ -3/4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

vi) **Find the mean square error,**

$$E = e_1^2 + e_2^2 + \cdots + e_n^2.$$

Mean Square Error, MSE $= (\frac{1}{4})^2 + (\frac{1}{4})^2 + (\frac{1}{4})^2 + (-\frac{3}{4})^2 = \frac{3}{4}$.

Extra: Using Calculus to Determine $\hat{x}$. Let, y and $\hat{y}$ denote the actual and predicted values.

Error Function,E $= \sum(y - \hat{y})^2 = \sum(y - (m\hat{x} + c))^2$
So, here, $E = (12 - 3m - c)^2 + (10 - 2m - c)^2 + (14 - 4m - c)^2 + (11 - 3m - c)^2$

Partial Derivative with respect to m,

$$\frac{\partial E}{\partial m} = 2(12-3m-c)(-3)+2(10-2m-c)(-2)+2(14-4m-c)(-4)+2(11-3m-c)(-3)$$

$$\frac{\partial E}{\partial m} = -6(12 - 3m - c) - 4(10 - 2m - c) - 8(14 - 4m - c) - 6(11 - 3m - c)$$

$$\implies \frac{\partial E}{\partial m} = (-72+18m+6c)+(-40+8m+4c)+(-112+32m+8c)+(-66+18m+6c)$$

$$\implies 0 = 76m + 24c - 290 \tag{5}$$

[To minimize E, $\frac{\partial E}{\partial m}$ is set to 0.]

Partial Derivative with respect to c,

$$\frac{\partial E}{\partial c} = 2(12-3m-c)(-1)+2(10-2m-c)(-1)+2(14-4m-c)(-1)+2(11-3m-c)(-1)$$

$$\frac{\partial E}{\partial c} = -2(12-3m-c)-2(10-2m-c)-2(14-4m-c)-2(11-3m-c)$$

$$\implies \frac{\partial E}{\partial c} = (-24+6m+2c)+(-20+4m+2c)+(-28+8m+2c)+(-22+6m+2c)$$

$$\implies 0 = 24m+8c-94 \tag{6}$$

[To minimize E, $\frac{\partial E}{\partial c}$ is set to 0.]

Solving equation(5) and equation (6) we get,$\hat{x} = (m, c) = (2, 23/4)$
Check this one out - Colab File

The Not-So-Special One's Masterplan: Boots vs Burgers

Ahead of the new season, the manager of Pulse FC, Mr. Not-So-Special One, decided to adopt a new strategy for his team. Often during the summer, players indulge in burgers, which negatively impacts their performance. This season, Mr. Not-So-Special One only wants elite match-winners on his team. According to him, players who score **more than** 15 goals in a season are considered elites.

In his quest to build a team of elite performers, he collected last preseason's stats, analyzing how many goals top strikers scored after devouring a good number of burgers during the summer holidays. Backed by these numbers, he's ready to assemble an undefeatable team for the season ahead.

Player Name	**Burgers Eaten**	**Goals Scored**
Romelu Lukaku	10	13
Eden Hazard	8	18
Gonzalo Higuaín	12	10
Harry Maguire	17	5

Table 2: Preseason Statistics

Player Name	**Burgers Eaten**	**Predicted Goals**	**Selection Status**
Asif	5		
Ananta	10		
Tanbir	4		
Tamim	8		

Table 3: Selection Table

Table 4: Comparison of Preseason Goals and Burger Consumption

Help Mr. Not-So-Special One to answer the following questions,

i) Write down the coefficient matrix A and solution vector b.

ii) Write down $\hat{x}$ in terms of matrix A and vector b.

$$\hat{x} =$$

iii) Find out the equation of the best fit line that represents the points in Table 2 using the answer of Question 1.b)

Answer : $goals_{predicted} =$ _____ $\times$ burgers $+$ _____

iv) Determine residual vector, $e = \begin{bmatrix} e_1 \\ e2 \\ e3 \\ e_4 \end{bmatrix}$

v) Based on the equation obtained in Question 1.c), complete Table 3. In the Predicted Goals column, write down the number of goals each player is expected to score in the upcoming season. Then, based on these predictions, fill in the Selection Status column with a Tick or 'Yes' if the player is selected and a Cross or 'No' if they are not.

Solution: SWE'22 Quiz 4

Better Luck Next Time Mr. Strang

The linear algebra course teacher of SWE 22, Professor Struggle, had been puzzled by the varying performance patterns in his class. He suspected there might be a direct relationship between how many hours students slept before their quizzes and the marks they achieved. To test his hypothesis, he meticulously tracked the sleep hours and corresponding quiz scores for five students over the semester. The results were compiled in the following Table 5.

Student	**Sleep Hours (x)**	**Quiz Marks (y)**
Mahbub	0	15
Awsaf	4	12
Mr. Strang	2	3
Pias	8	13
Utsho	16	9

Table 5: Table Representing the Correlation Between Sleep Duration and Quiz Scores of Students

Answer the following questions,

i) An outlier is a data point that significantly deviates from the overall pattern in a dataset, often lying far from the regression line. Identify the outlier(based on intuition) among the five given points in Table 5 and briefly justify why it qualifies as an outlier. (Exclude the outlier from any calculations in the following questions.)

ii) Using matrix algebra, determine the equation of the best-fit straight line, $y = mx + c$, for the four points (excluding the outlier) in Table 5. Also, find the error vector, e.

iii) Determine the equation of the best-fit quadratic curve, $y = ax^2 + bx + c$, for the four points (excluding the outlier) in Table 5 using matrix algebra. Also, determine the residual vector (error vector), e, for this case.

iv) Which of the two representations obtained in item ii) and item iii) provides a better fit to the data and why?

v) If a new data point is introduced where a student who slept for 10 hours scored 10 marks on the quiz, which of the two equations obtained in item ii) and item iii) predicts the quiz score more accurately? Does this contradict your answer in item iv)?

Solution: SWE'22 Final Question 3